

ISIT312 Big Data Management

Extraction, Transformation, and Loading

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Extraction, Transformation, and Loading

Outline

Extraction, Transformation, and Loading

Conceptual ETL Design using BPMN

Conceptual Design of the Northwind ETL

Extraction, Transformation, and Loading (ETL)

Extract data from internal and external sources, **transform** data, and **load** data into a data warehouse (**ETL**)

No agreed way to specify **ETL** at a conceptual level

We study conceptual **ETL** design

Conceptual model based on the **Business Process Modeling Notation (BPMN)**

- Users already familiar with **BPMN** do not need to learn another language to design **ETL**
- **BPMN** provides a conceptual and implementation-independent specification of processes
- Processes expressed in **BPMN** can be translated into executable specifications(e.g., Microsoft's Integration Services)

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Conceptual ETL Design using BPMN

Basic assumption for using **BPMN** as conceptual model: **ETL** process is a type of business process

There is no standard model for defining **ETL** processes

Each tool provides its own model, too detailed to be conceptual

Using **BPMN** constructs we define the most common **ETL** tasks and define a **BPMN** notation for **ETL**

ETL process: A combination of **control** and **data processes**

- **Control processes** manage the coarse-grained groups of tasks
- **Data processes** detail how input data are transformed and output data are produced

Two kinds of tasks in **ETL** conceptual modeling

- **Control tasks** highlight the control procedures provided by **BPMN**. Represent a **workflow** (arrows represent the precedence between activities)
- **Data tasks** refer to the tasks that directly manipulate data during an ETL process. Represent a **data flow** (arrows represent data 'flowing' along them)

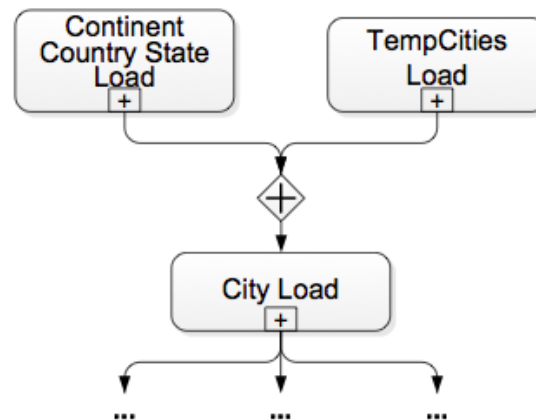
Control Tasks

Represent the workflow sequence or **orchestration** of the **ETL** process independently of the data flow

Control tasks are represented by means of **BPMN** constructs

For example, gateways are used to control the sequence of activities in an **ETL** process

The most used types of gateways in an **ETL** context are exclusive and parallel



Data Tasks

Show how data are manipulated **within** an activity

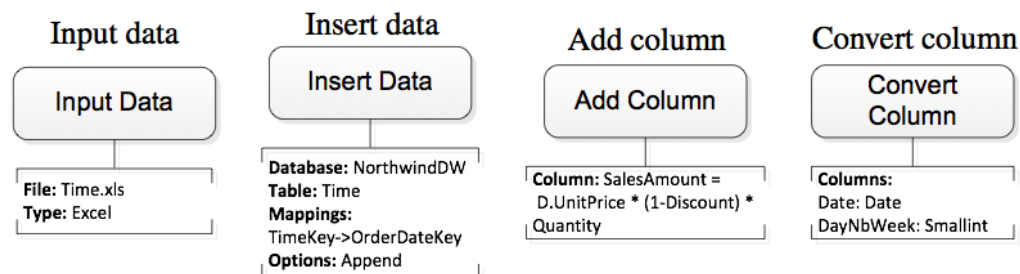
At lower abstraction level than control tasks

Represent activities typically carried out to manipulate data: input and output data, data conversion and transformation, for instance, change the data type of an attribute, add a column, remove duplicates, and so on

We denote these tasks **unary data tasks** since they receive one input flow

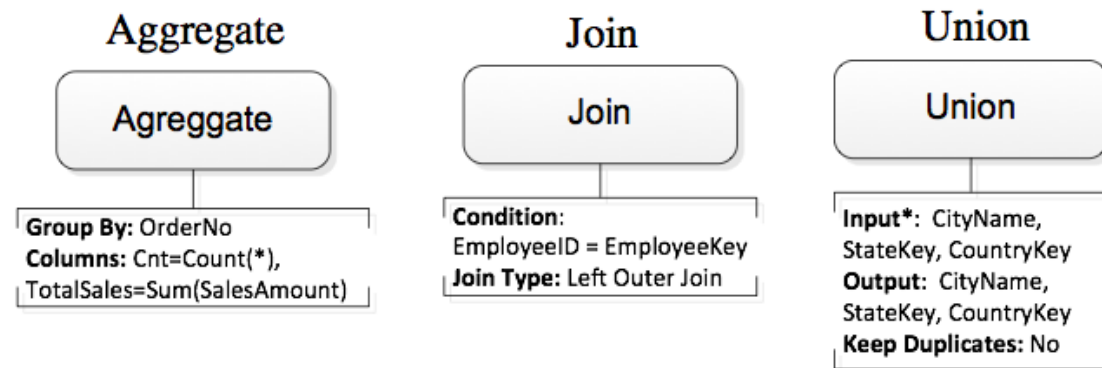
n-ary data tasks receive as input more than one flow (e.g., this is the case of union, join, difference,...)

Row operations are the transformations applied to the source or target data on a row-by-row basis, e.g., updating the value of a column



Rowset Data Tasks

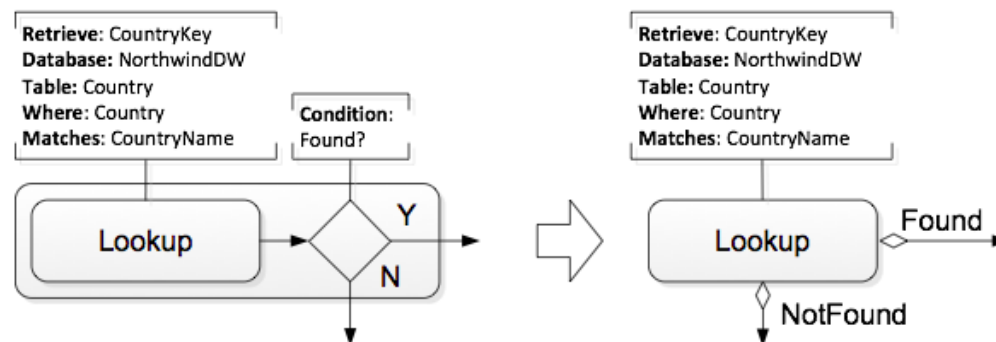
Rowset operations deal with a set of rows, e.g., aggregation is a rowset operation



Lookup Data Tasks

Lookup Data Tasks check if some value is present in a file. Immediately followed by an exclusive gateway with a branching condition. We use a shorthand replacing these two tasks by 2 conditional flows.

Shorthand notation for the lookup task



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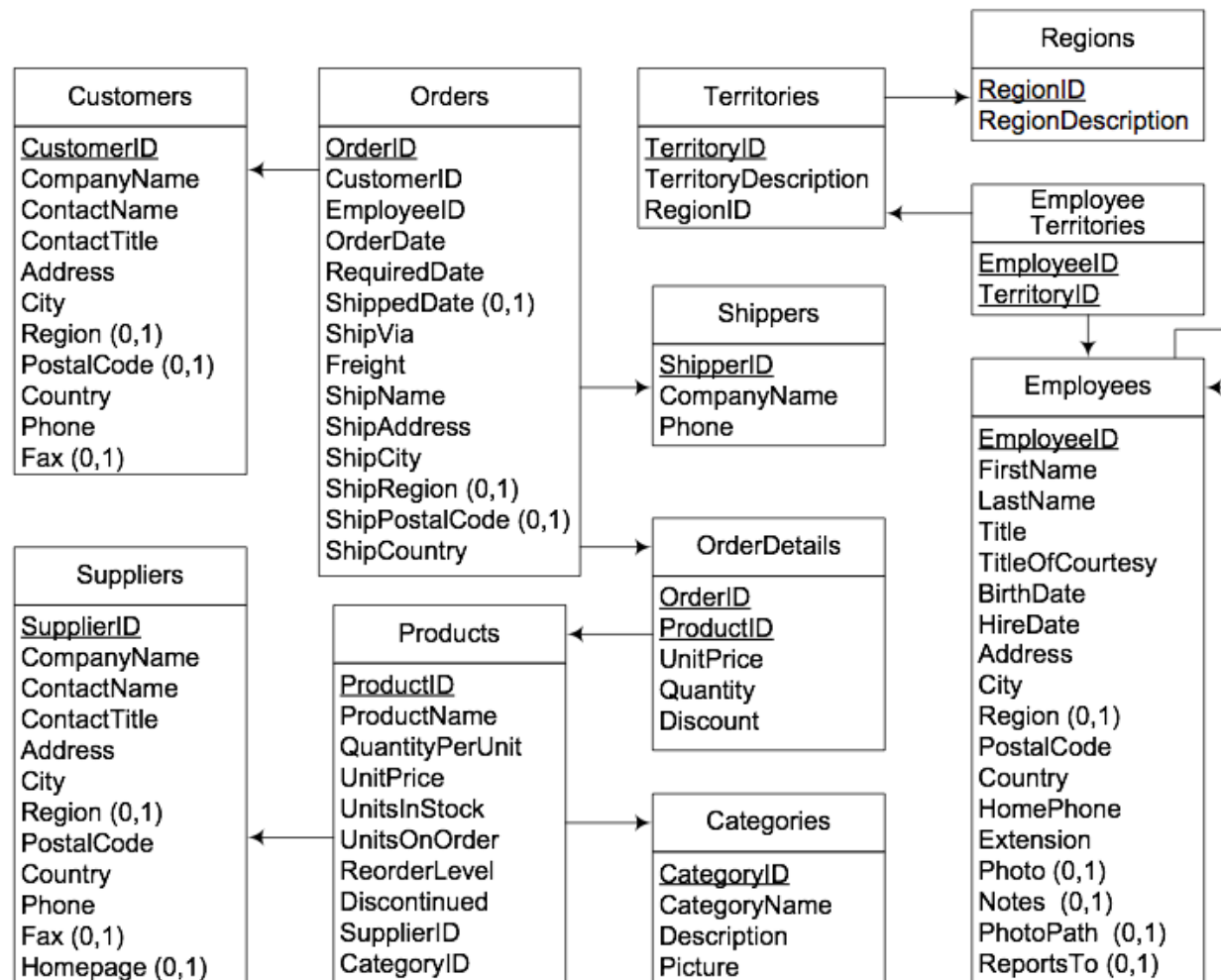
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Schema of the Northwind Operational Database



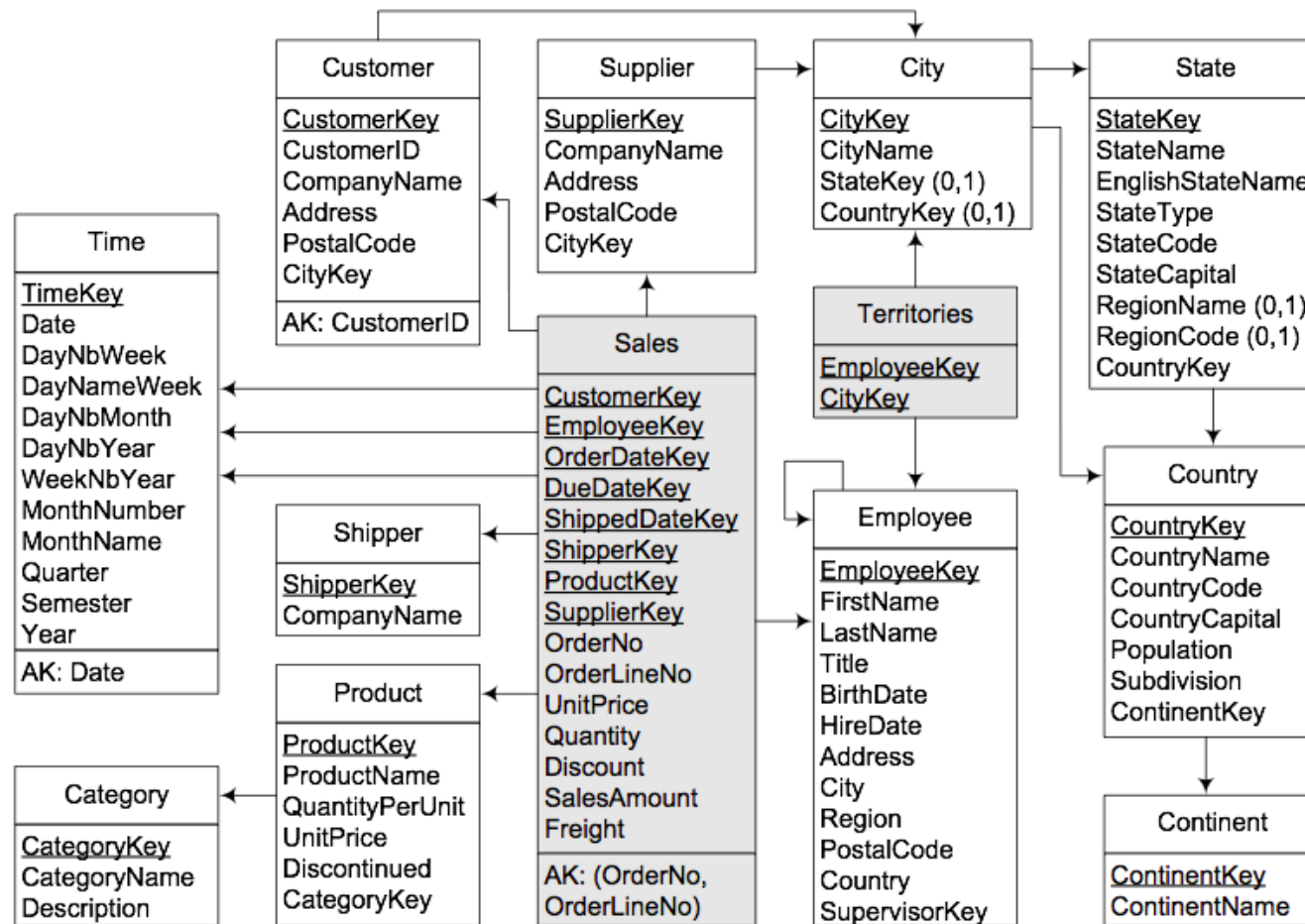
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Schema of the Northwind Data Warehouse



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Conceptual Design of the Northwind ETL: Data Sources

File **Time.xls** contains data for loading the **Time** dimension, spanning the dates in table **Orders** of the operational database

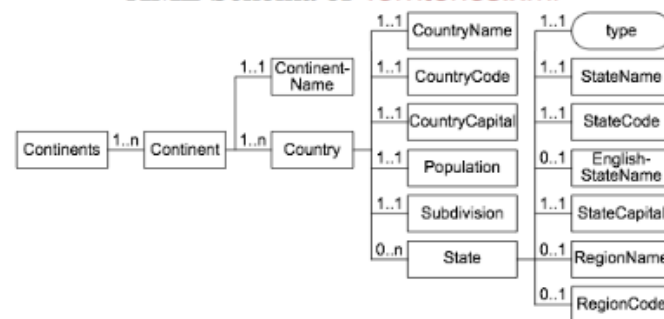
Dimensions **Customer** and **Supplier** share the geographic hierarchy starting at the **City** level

Data for the hierarchy **State** → **Country** → **Continent** loaded from **Territories.xml**

Start of the file **Territories.xml**

```
<?xml version="1.0" encoding="ISO-8859-1"?>
<Continents>
  <Continent>
    <ContinentName>Europe</ContinentName>
    <Country>
      <CountryName>Austria</CountryName>
      <CountryCode>AT</CountryCode>
      <CountryCapital>Vienna</CountryCapital>
      <Population>8316487</Population>
      <Subdivision>Austria is divided into nine Bundeslnder,
        or simply Lnder (states; sing. Land).</Subdivision>
      <State type="state">
        <StateName>Burgenland</StateName>
        <StateCode>BU</StateCode>
        <StateCapital>Eisenstadt</StateCapital>
      </State>
      <State type="state">
        <StateName>Krnnten</StateName>
        <StateCode>KA</StateCode>
        <EnglishStateName>Carinthia</EnglishStateName>
        <StateCapital>Klagenfurt</StateCapital>
      </State>
    </Country>
  </Continent>
  ...
</Continents>
```

XML Schema of **Territories.xml**



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Conceptual Design of the Northwind ETL: Data Sources

File called **Cities.txt** identifies to which state or province a city belongs

Contains three fields separated by tabs and begins as shown below

For cities located in countries that do not have states (e.g. Singapore), second field is set to null

The file is also used to identify to which state corresponds the city in the attribute TerritoryDescription of table **Territories**

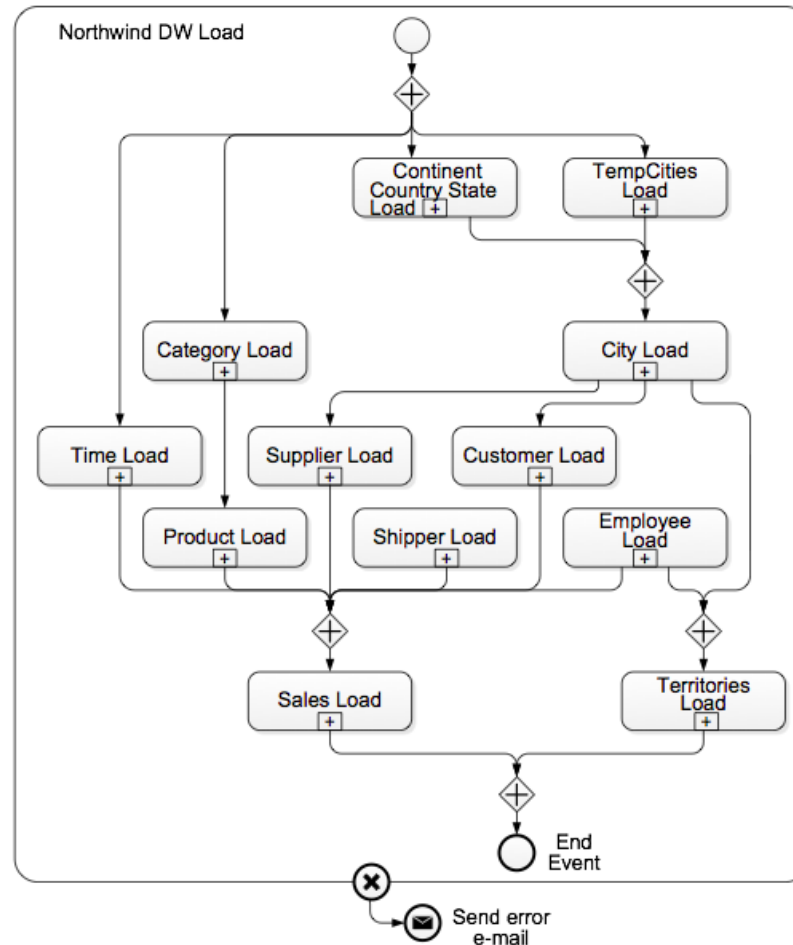
City → State → Country Aachen → North Rhine-Westphalia → Germany Albuquerque → New Mexico → USA Anchorage → Alaska → USA Ann Arbor → Michigan → USA Annecy → Haute-Savoie → France ...
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Beginning of the file **Cities.txt**

TempCities
City
State
Country

Associated table **TempCities**

Conceptual Design of the Northwind ETL: Overall View



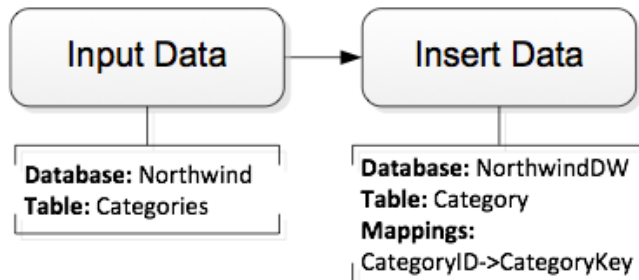
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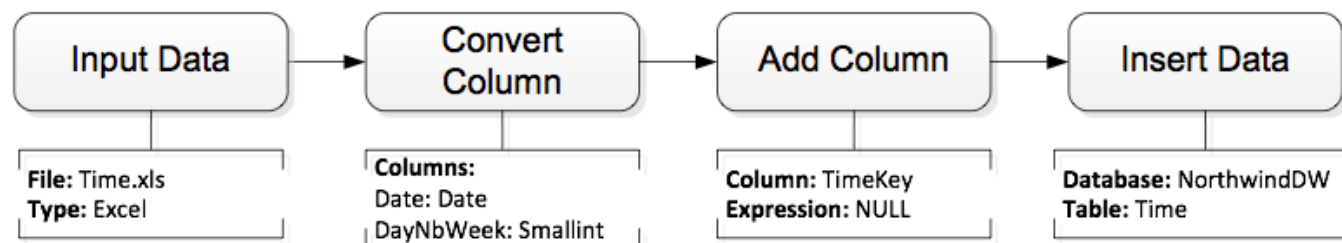
Conceptual Design of the Northwind ETL

Load of the **Category** dimension table



- Input task loads table **Categories** from the operational database
- Insert task loads the table **Category** in the data warehouse, mapping **CategoryID** to **CategoryKey** attribute in the **Category** table

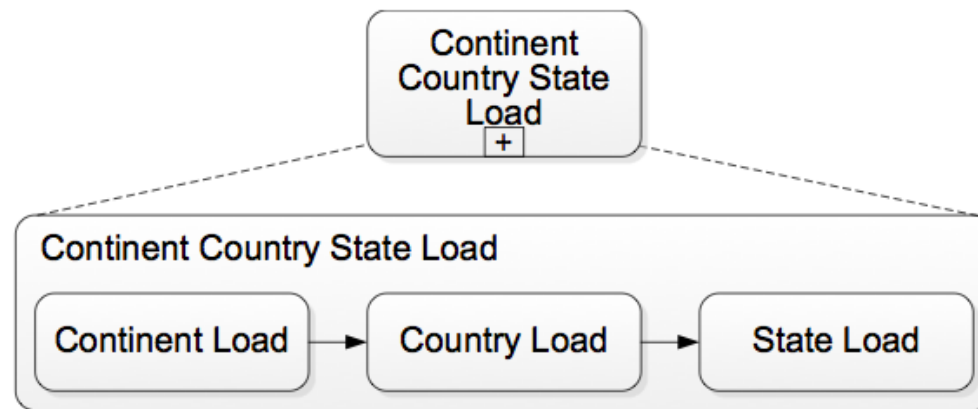
Loading the **Time** dimension table from an Excel file is similar, but includes a data type conversion, and an addition of the column **TimeKey**



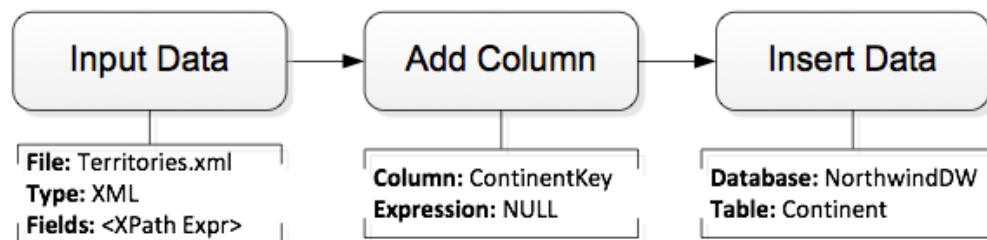
Conceptual Design of the Northwind ETL

Loading the **City** level first requires loading the **Geography** hierarchy
State → **Country** → **Continent**

Associated control task



Load of the **Continent** table



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References

A. VAISMAN, E. ZIMANYI, Data Warehouse Systems: Design and Implementation, Chapter 8 Extraction, Transformation, and Loading, Springer Verlag, 2014